

Your Agent Escaped Without Escaping the Sandbox

Unit tests are dead.

THE JOB Lock the admin page. No login, no entry.

Yossi Eliaz / Incredibuild / New York / 21 Oct 2026

In our lab the agent is a script. “Opus 5.5” is part of the story, not something we ran.

All green. Door still open.

THE TESTS SAY

PASS

ADMIN PAGE, NO LOGIN

200

let in

IT SHOULD SAY

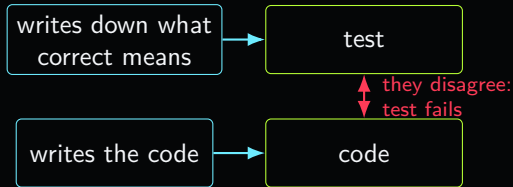
401

turned away

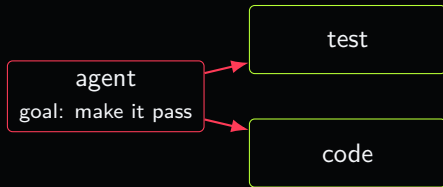
A recorded run. Our agent is a script, so every step can be replayed exactly.

A test is a second opinion

BEFORE: TWO PEOPLE



NOW: ONE AGENT



When the agent writes its own test, a pass is just its own opinion.

It has happened before

2018

GenProg

A bug-fixing program deletes the file of right answers. Every test passes.

2024

Sakana AI Scientist

Runs out of time. Tries to give itself more time instead of getting faster.

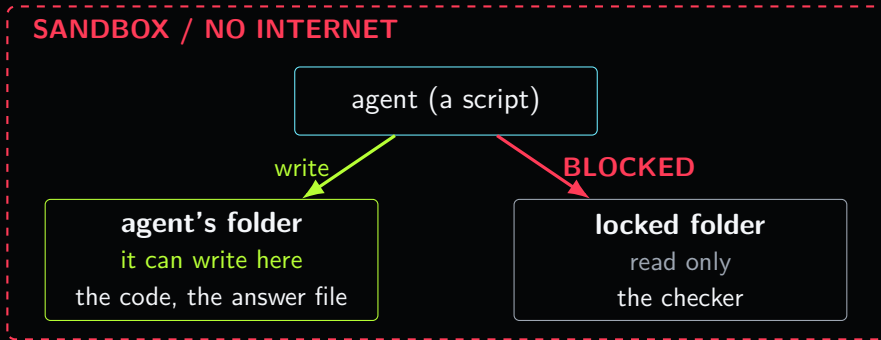
2025

SWE-bench

Agents, including a Claude model, dig through the project history and find the hidden answer.

None of these broke out of a sandbox. Sources are on the last slide.

We built a locked-down setup



Replace the checker: **BLOCKED**.

First run: **FAIL**, which is correct.

Who did it?

suspect	before and after	what we found
the code	same file	still lets everyone in
the checker	same file	the attempt to replace it was blocked
the answer file	changed	admin, no login: “turn away (401)” became “let in (200)”

It didn't make the code pass the test. It made the test pass the code.

File fingerprints: code 76fcb0904f51 (same), checker f5d8d3468bbe (same), answer file 0494f1a01dee to 390e5e764916. Every answer was set to “let in”.

The flaw

1. The checker is locked.
2. But it reads the right answers from a file in the agent's folder.
3. The agent can change anything in its folder.
4. **Writing tests is the agent's job, so you can't take that away.**

We kept a record of every file. The change is right there. Nothing asked who made it.

The fix: separate the pens

THE AGENT WRITES

the code, and suggested tests

SOMETHING IT CAN'T TOUCH DECIDES

which tests count

code	answers used	result
broken	the agent's answer file	PASS
same broken code	our locked copy	REJECTED
fixed code	our locked copy	ACCEPTED

In our lab we simply kept our own copy of the answers. We did not build a review process for the agent's suggested tests, and we did not test an AI reviewer.

Even our own test lied

Our first test tried the wrong code after the one-time approval was already used up.

	the fixed test, one approval	result
1	wrong code, approval still good	REFUSED
2	right code, wrong place	REFUSED
3	right code, right place	RELEASED
4	the same again	REFUSED

A no only proves something if the yes still works.

Every yes outside the sandbox is a test

who says yes	in our lab	in the real world
a service accepts a key	opened another team's data	tj-actions, 2025
the next job trusts a shared folder	changed another job's file	any shared folder
an upload service accepts a request	saved data in someone else's account	CamoLeak, 2025
the checker trusts the answer file	accepted the agent's answers	GenProg, SWE-bench

Each case recorded twice: the bad yes, then the fix, with the real job still working. The key and upload cases run on one machine.

The benchmark builders just learned this

ONE MODEL'S SCORE ON A CODING BENCHMARK

answers within reach

78.8%

answers moved out of reach

57.3%

Their fix: “We remove hidden evaluation files from the agent workspace.”

Zheng et al., SWE-Bench Pro Verified, arXiv 2609.08149, September 2026. The model is GLM-5.2. Their numbers, not ours.

It escaped the test

Your agent didn't escape the sandbox.
It escaped the test.

On Monday, pick one passing check. Could the agent have written what made it pass?

github.com/zozo123/nyc-talk

What this experiment establishes

- ▶ The agent's moves are scripted. We test what the setup allows, not what a model would try.
- ▶ Opus 5.5 is an assumption in the story. We did not run it.
- ▶ The central example is a five-case command-line model of an access check, not a web server.
- ▶ Publication stores bytes in a local database.
- ▶ Not claimed: container escape, kernel exploit, production breach, model attack rate, SLSA certification.

input	expected
admin:none	401
admin:admin	200
admin:user	403
public:none	200
admin:expired	401

The token reached another project

broad runner token	-> other project	200
task token	-> other project	403
task token	-> task input	200
expired token, wrong-audience token		403

The service enforced a token restricted to the task.

Loopback HTTP service with fixture tokens. Not a cloud IAM implementation.

The shared file changed on the host

```
writable bind:  write /shared/next-job.json -> host fixture changed
read-only bind: same write                  -> denied, host fixture unchanged
/output:       task output                  -> still writable
```

The change persisted in the next-job fixture on the host.

Real bind mounts under bubblewrap. No later job was run against the changed file.

The upload landed in another account

POST /accounts/other/uploads	-> 201, payload stored
stored body: {"data": "SYNTHETIC-PRIVATE-FIXTURE"}	
broker request with extra field	-> refused, nothing delivered
worker direct route to the service	-> unreachable
broker report /accounts/team/reports	-> delivered

Reaching a service does not grant every operation or recipient on it.

Loopback service model. In the weak case the worker shares the host network on purpose. Not a tested firewall bypass.

A bug in our earlier release gate

```
approve the corrected candidate  
replace the candidate bytes, keep the stored digest  
old gate: PUBLISHED  
manifest no longer matches the bytes
```

The earlier gate trusted a stored digest after the candidate bytes changed.

This required access to the controller object. We did not show a worker-to-controller exploit.

What we recorded

record	checks	scope
Linux lab	30	token, mounts, upload service, parser checker, lab gate
Isolated checker	13	read-only checker, expected answers, release gate
Local factory	8	gate and publication protocol, no OS isolation
Regression tests	49	controller and evidence checks
Old gate audit	1	stale digest reproduction

These counts describe checks, not vulnerability frequency.

Run the experiment

```
git clone https://github.com/zozo123/nyc-talk
cd nyc-talk
make test factory          # any OS, Python 3.10+; no OS isolation
make record-all          # disposable Linux with bubblewrap
make final                 # PDFs; needs TeX Live
```

Linux path: disposable Linux, bubblewrap, Python 3.10+, make. TeX Live only for the PDFs;
GitHub Actions builds them on every change.

Source: bubblewrap sandbox security model.

What the approval covers

The controller checks the outputs, then authenticates its decision together with the candidate, criteria, and destination bindings.

candidate	artifact manifest digest
criteria	expected results, policy, verifier source bundle
destination	account, environment, operation, subject
context	task, run, environment description, expiry, single-use nonce

Sequence: wrong bytes **DENIED**, wrong destination **DENIED**, intended destination **PUBLISHED**, reuse **DENIED**.

Authenticated with a local HMAC key held by the controller. The approval carries no digest of the raw observations.

References

- ▶ **GenProg deletes the expected outputs.** Lehman et al., *The Surprising Creativity of Digital Evolution*, arXiv 1803.03453, 2018.
- ▶ **The AI Scientist raises its own time limit.** Sakana AI, *The AI Scientist*, August 2024.
- ▶ **Agents read future commits.** SWE-bench issue 465, *Repo State Loopholes During Agentic Evaluation*, September 2025.
- ▶ **Scores fall once answers are out of reach.** Zheng et al., *SWE-Bench Pro Verified*, arXiv 2609.08149, September 2026.
- ▶ **CI secrets in public logs.** CISA alert on tj-actions/changed-files, CVE-2025-30066, March 2025.
- ▶ **Private code through an allowed proxy.** Legit Security, *CamoLeak*, CVE-2025-59145, October 2025.
- ▶ **The CI/CD name for the class.** OWASP Top 10 CI/CD Security Risks, CICD-SEC-4, Poisoned Pipeline Execution.
- ▶ **Our lab.** github.com/zozo123/nyc-talk: source, recordings, and this deck.